

- 4 The area of a triangle is  $12 \text{ cm}^2$ .  
The length of the base of the triangle is  $8 \text{ cm}$ .

Work out the height of the triangle.

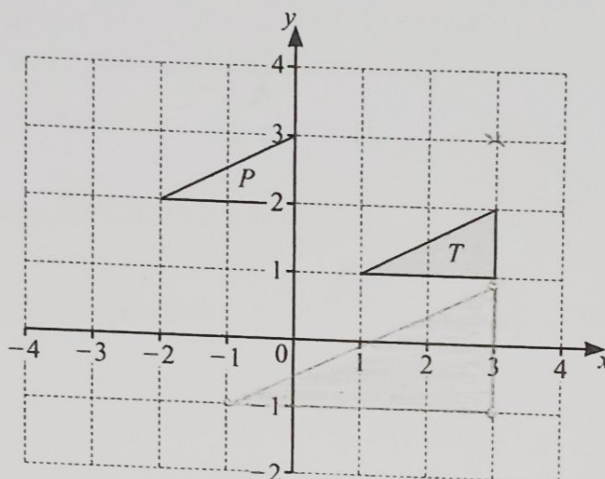
$$= \frac{1}{2} \times b \times h \quad h = 3 \text{ cm}$$

$$12 = \frac{1}{2} \times 8 \times h$$

$$\frac{24}{8} = \frac{8h}{8}$$

5

..... 3 ..... cm [2]



$$A = \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$$

$$C = \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$$

- (a) Describe fully the **single** transformation that maps triangle  $T$  onto triangle  $P$ .

Translate triangle T by the vector  $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$

- (b) Draw the image of triangle  $T$  after an enlargement of scale factor 2, centre  $(3, 3)$ .

- 6 Find the value of

(a)  $5^{-5} \times 5^5$

(b)  $125^{\frac{2}{3}}$

$$\left( \sqrt[3]{125} \right)^2 = (5)^2 = 25$$

..... 1 ..... [1]

..... 25 ..... [2]